

Lab 1: Data Masking

- **Objective:** Implement a "Last-Four" masking rule to protect PII (Personally Identifiable Information) while maintaining enough data for customer verification.
- **Scenario:** You are a Junior Developer at **SecureBank**. Your manager has asked for a utility script. When a customer calls support, the agent should only see the last 4 digits of the card to confirm they are talking to the right person.
 - **Input String:** "4532-8812-9001-4321"
 - **Target Output:** "XXXX-XXXX-XXXX-4321"
- Step 1: Breakdown
 - Before you can mask the data, you need to isolate the parts that stay and the parts that go.
 - create a variable and use the `.split()` method on the hyphen (-).
 - print variable, it should look like a list of four separate strings
- Step 2: Loop
 - Now, iterate through your list.
 - you can loop with a -1 index
- Step 3: Re-assembly
 - A list of strings isn't helpful for a display UI. You need to turn ['XXXX', 'XXXX', 'XXXX', '4321'] back into a single string.
 - Use the `"-".join()` method to combine them.
- Constraints
 - **Constraint A:** Your code must work regardless of whether the digits are X or *.
 - **Constraint B:** Your code must preserve the hyphens. If the output is XXXXXXXXXXXXX4321, it is incomplete
- Validation Test
 - Once your code works for the credit card, test it with this **Phone Number**:
 - "123-456-7890"
 - If your function is written well, it should automatically output
 - "XXX-XXX-7890"
- Individual
 - What to submit
 - Source code
 - Documentation (screenshots of results as well as short descriptions) in .docx/.pdf